
Information Technology University

Department of Computer Science / Artificial Intelligence

Machine Learning
Semester Project Proposal

Skin Disease Classification

Submitted By	
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1. Introduction and Background

Skin diseases are very common and affect around 900 million people worldwide. Some of these conditions can be life-threatening if not caught early. Detecting them early and correctly is one of the best ways to help patients recover and reduce treatment costs.

Doctors usually diagnose skin diseases by looking at the skin directly or using a tool called a dermoscope, which uses special light to see below the skin surface. This method depends a lot on the doctor's experience, takes time, and is not always available, especially in remote areas. In Pakistan, there are very few skin specialists, and many people in rural areas cannot access proper skin care at all.

We will use the **Skin diseases image dataset** from kaggle by Ismail Hossain.

2. Problem Description and Project Objective

2.1 Problem Description

Classifying skin diseases from images is difficult for three main reasons.

- **Similar appearance:** Many skin diseases look very similar to each other in images, which makes it hard for a model to tell them apart accurately.
- **Unbalanced data:** Some disease types have thousands of images while others have very few. This causes the model to get good at recognizing common diseases but miss the rare ones, which are often the most dangerous, like melanoma.
- **Complex patterns:** Older machine learning methods that rely on manually picked features are not good enough to pick up the small differences in texture and color that separate one skin condition from another.

Because of these problems, we need a learning model that can handle unbalanced data, process images properly, and perform well across all disease types, not just the common ones.

2.2 Project Objective

To build and test a CNN model from scratch that can automatically identify the type of skin disease from a dermoscopic image. The model will be trained on the **Skin diseases image dataset**. We will use techniques to handle the imbalance between disease classes, and will be compared against simpler baseline models to measure how much improvement deep learning provides.

Metric	Target	What it Means
Overall Accuracy	> 85%	The model correctly identifies the disease most of the time
Macro F1-Score	> 0.80	The model performs well across all 10 disease types

3. System Design

The system will consist of several components that work together.

1. **Data Collection** : The dataset is downloaded from Kaggle (Skin diseases image dataset).
2. **Data Preprocessing** : Images are resized, pixel values are normalized, and random changes like flips and rotations are applied to create more training variety.
3. **Model Training** : We train our CNN models on the prepared images. We start with simple baseline models, then move to our custom deep CNN.
4. **Evaluation** : We measure how well the model performs using accuracy, F1-score, etc across all 10 disease types.
5. **Interface / Output** : A simple web app lets users upload a skin image and see which disease the model predicts, along with a confidence score.

4. Our Approach and Plan

The project will follow these steps:

- **Data Gathering**: Download the Skin diseases image dataset from Kaggle.
- **Data Exploration (EDA)**: Look at how many images exist per disease type, visualize samples, and check for quality issues using graphs and charts.
- **Data Preprocessing**: Resize all images to 224x224 pixels, normalize pixel values, and apply augmentation (flips, rotations, color changes) to create more variety especially for rare disease types.
- **Baseline Model** : First train a simple model (Logistic Regression or a basic 3-layer CNN) to set a starting performance level to compare against.
- **Model Evaluation** : Compare results using accuracy, F1-score, etc.

Scope: The project focuses on model building and evaluation. We will not use any pre-trained model weights, deploy to mobile, or work with non-standard image types.

Ethics: All images are publicly available and contain no personal information. The dataset is free to use for research. Our tool is meant to assist doctors, not replace them.

5. Dataset

We will use the Skin diseases image dataset, a publicly available dataset on Kaggle with 27,153 images covering 10 different skin disease types. It is suitable for training a multi-class image classification model from scratch.

Property	Details
Dataset Name	Skin Diseases Image Dataset
Source	Kaggle: kaggle.com/datasets/ismailpromus/skin-diseases-image-dataset
Total Images	27,153 labeled images
Disease Types	10 classes
Image Type	Color (RGB), JPEG format
License	Publicly available for research use

5.1 Disease Types in the Dataset

#	Disease Name	Description
1	Cellulitis	Bacterial skin infection causing redness and swelling
2	Impetigo	Contagious bacterial infection with blistering rash
3	Athlete Foot	Fungal infection between the toes
4	Nail Fungus	Fungal infection affecting nails
5	Ringworm	Circular fungal rash on skin
6	Chickenpox	Viral infection causing itchy blistering rash
7	Cutaneous Larva Migrans	Skin infection from parasitic larvae
8	Eczema	Chronic inflammatory skin condition causing dryness and itching
9	Psoriasis	Autoimmune condition causing scaly skin patches
10	Warts / Molluscum	Viral skin growths

Note: Some disease types may have more images than others. We will handle any imbalance using weighted loss functions and data augmentation during training.

6. Tools and Technologies We Will Use

Tool / Library	What We Use It For
Python	Main programming language
PyTorch 2.x	Building and training our CNN model

React/Next Js	Simple web app to upload images and see predictions
Google Colab	Free online GPU to speed up model training
GitHub	Saving and sharing code between team members

7. Team & Responsibilities

Team	
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Responsibilities TBD.